Quantum Computing and Option Pricing

Mnacho Echenim

Grenoble INP - Ensimag, UGA

2025-2026

Organisation générale du cours

- Déroulé des séances
  - Séances de 1h20
    - **On commence à l'heure**
  - 4 CM (lundi 8h15) et 4 TD (mercredi 11h15)
  - 1 Permanence
  - 3 séances **encadrées** de projet
- Transparents
  - Moins de temps de recopie, plus de temps d'**interaction**
  - Transparents à trous pour laisser décanter les concepts
  - **Transparents en anglais**
- Fiche d'exercice avec les solutions

Outline

On quantum computation and information

- A blooming field with major public and private actors
  - Over 25 billion dollars of public investment in 2021 (McKinsey), probably over 40 billion dollars in the past decade
  - Big tech companies: Google, IBM, Amazon, Microsoft
  - A large range of quantum technology startups (IonQ, Rigetti Computing, Alice & Bob...), including in Grenoble: `https://quobly.io/`

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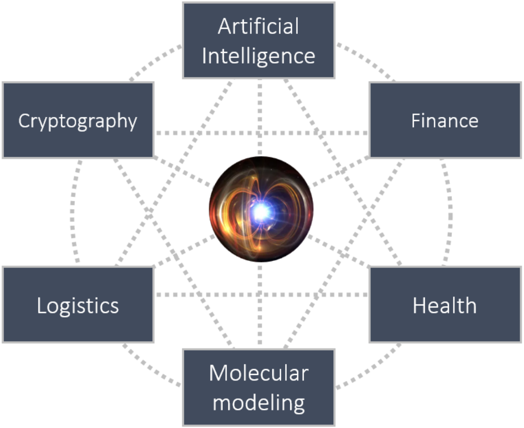
- Still a large research playground
  - Over 430 algorithms are registered on the Quantum Algorithm Zoo `https://quantumalgorithmzoo.org/`
  - Research on designing algorithms, on proving they are correct, on exploring quantum advantages...

On quantum computation and information (cont'd)



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  - Main postulates
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- What you will know at the end of the class
  - The quantum mechanics postulates and some of their implications
  - How to design basic quantum circuits
  - How to test the correctness of a quantum circuit
  - How to implement a quantum algorithm using Qiskit
  - Some of the potential and limits of quantum information and computation

Timetable

Lecture 1 Postulates of quantum mechanics

- Quantum state representation
- Quantum state evolution
- Quantum state measurement
- *Application: superdense coding*

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Hands-on A quantum circuit for pricing a call option with Qiskit

Resources

- Books
  - M. Burniat, T. Damour: *Le mystère du monde quantique*
    (https://www.dargaud.com/bd/le-mystere-du-monde-quantique-bda5133660)
  - O. Ezratty: *Comprendre l'Informatique Quantique*
    (https://www.oezratty.net/wordpress/2020/comprendre-informatique-quantique-edition-2020/)
  - M. Nielsen, I. Chuang: *Quantum Computation and Quantum Information*
    (http://mmrc.amss.cas.cn/tlb/201702/W020170224608149940643.pdf)
  - R. de Wolf: Quantum Computing: *Lecture Notes*
    (https://homepages.cwi.nl/~rdewolf/qcnotes.pdf)
  - J. Watrous: Understanding Quantum Information and Computation (Unit 1)
    (https://jhwatrous.github.io/UQIC.pdf)
- Videos
  - F. Magniez: *Cours au Collège de France*
    (https://www.college-de-france.fr/site/frederic-magniez/course-2020-2021.htm)

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Some final thoughts:

John Wheeler If you are not completely confused by quantum mechanics, you do not understand it

Richard Feynman I think I can safely say that nobody understands quantum mechanics

Roger Penrose Quantum mechanics makes absolutely no sense

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The qubit

- The simplest quantum system is the qubit
- The state of a qubit is described by a dimensional vector $\vec{\psi}$ in $\mathbb{C}^2$ with norm 1: $\|\vec{\psi}\| = 1$

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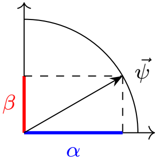

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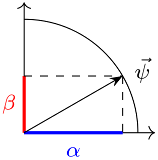

Notation

Instead of writing $\vec{\psi} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ where $|\alpha|^2 + |\beta|^2 = 1$, we will write

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

where $|0\rangle \stackrel{\text{def}}{=} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle \stackrel{\text{def}}{=} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

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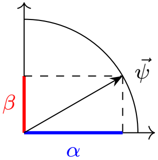

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Remark

The notation above is called the **Dirac notation**, we pronounce "ket ψ"

Ket and bra notations

Definition

If $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ then we let $\langle\psi| \stackrel{\text{def}}{=} (\overline{\alpha} \quad \overline{\beta})$

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Proposition

If $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $|\varphi\rangle = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}$ then

$\langle\psi| \cdot |\varphi\rangle = (\overline{\alpha} \quad \overline{\beta}) \cdot \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \overline{\alpha} \cdot \alpha' + \overline{\beta} \cdot \beta' = \langle\psi|\varphi\rangle$, the inner product of $|\psi\rangle$ and $|\varphi\rangle$

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A graphical representation

- A qubit represented by $|\psi\rangle$ is characterized by two complex numbers
  - It can be represented by 4 real numbers, but
  - The fact that $|\psi\rangle$ has norm 1 creates a constraint and makes it possible to write (up to a global phase)

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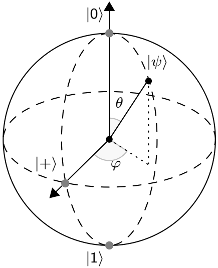
- A qubit can be represented on the Bloch sphere



Transforming qubits

Postulate (unitary evolution)

A qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ can be transformed by any function U that is

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Remarks

- The image of $|\psi\rangle$ by an operation will always be represented as $U \cdot |\psi\rangle$ for some matrix U
- All quantum transformations are invertible, all transformations of a quantum system can therefore be reversed

Examples of qubit transformations

- The NOT gate $X \overset{\text{def}}{=} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
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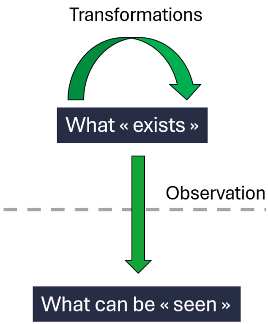

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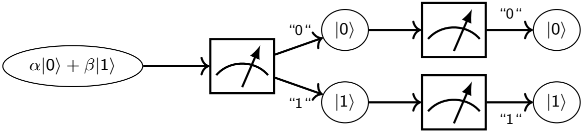

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Systems involving several qubits

- A qubit represents the simplest nontrivial quantum system
- In order to exploit the full potential of quantum mechanics, it will be necessary to handle several qubits simultaneously
- In what follows
    - Representation, tranformation and measurement of systems involving two qubits
    - Generalization to an arbitrary number of qubits

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We have $(|\psi_1\rangle \otimes |\psi_2\rangle)^\dagger \cdot (|\psi_3\rangle \otimes |\psi_4\rangle) = (\langle\psi_1| \otimes \langle\psi_2|) \cdot (|\psi_3\rangle \otimes |\psi_4\rangle) = \langle\psi_1|\psi_3\rangle \cdot \langle\psi_2|\psi_4\rangle$

Entanglement

- In the general case a quantum system involving two qubits is described by a vector in $(\mathbb{C}^2)^{\otimes 2} \simeq \mathbb{C}^4$ with norm 1
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Important note

Entanglement is a key property both in quantum computing and quantum cryptography

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  - This is an entangled state
  - It is called a **Bell state**

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Swapping both qubits: $\text{SWAP} \cdot (|a\rangle \otimes |b\rangle) = |b\rangle \otimes |a\rangle$

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Constructing an entangled state

Question

What transformation permits to generate the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ starting from the state $|00\rangle$?

Answer: $\text{CNOT} \cdot (H \otimes \mathbf{I})$

$$\begin{aligned}(H \otimes \mathbf{I}) \cdot |00\rangle &= H \cdot |0\rangle \otimes \mathbf{I} \cdot |0\rangle \\ &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle \\ &= \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\end{aligned}$$

$$\begin{aligned}\text{CNOT} \cdot (H \otimes \mathbf{I}) \cdot |00\rangle &= \text{CNOT} \cdot \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \\ &= \frac{1}{\sqrt{2}}\text{CNOT} \cdot (|00\rangle + |10\rangle) \\ &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\end{aligned}$$

Measuring two qubits

- Consider the qubit state $|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$
- Given $i, j \in \{0, 1\}$, if both qubits are measured simultaneously, then the output will be "ij" with probability $\left|\alpha_{ij}\right|^2$
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  - The output will be "1" with probability $P_1 = |\alpha_{10}|^2 + |\alpha_{11}|^2$ and the state post-measurement will be

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    $$\frac{1}{\sqrt{P_1}}(\alpha_{10}|10\rangle + \alpha_{11}|11\rangle)$$
- The statement when only the second qubit is measured can be transposed straightforwardly

Examples

- Measuring both qubits on $\frac{1}{3}|00\rangle + \frac{\sqrt{2}}{3}|01\rangle - \frac{1}{\sqrt{3}}|10\rangle - \frac{1}{\sqrt{3}}|11\rangle$
  - "00" is output with probability $\frac{1}{9}$ and "10" is output with probability $\frac{1}{3}$

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  - "1" is output with probability $\frac{1}{2}$ and the post-measurement state is $|11\rangle$
  - If the second qubit is measured, the output will be perfectly correlated with that of the first qubit
    - Einstein: "*spooky action at a distance*"

Generalization to an arbitrary number of qubits

States

A system involving n qubits is represented by a vector $|\psi\rangle \in (\mathbb{C}^2)^{\otimes n}$, we have

$$|\psi\rangle = \sum_{(b_0,\dots,b_{n-1})\in\mathbb{Z}_2^n} \alpha_{b_{n-1}\cdots b_0} |b_{n-1}\cdots b_0\rangle$$

where $|b_{n-1}\cdots b_0\rangle = |b_{n-1}\rangle \otimes \cdots \otimes |b_0\rangle$ and $\sum_{\vec{b}\in\mathbb{Z}_2^n} \left|\alpha_{\vec{b}}\right|^2 = 1$

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Transformations

The transformations on systems involving n qubits are exactly the set of unitary matrices over $\mathbb{C}^{2^n\times 2^n}$

Generalization to an arbitrary number of qubits (cont'd)

Measurements

Given $|\psi\rangle = \sum_{\vec{b}\in\mathbb{Z}_2^n} \alpha_{\vec{b}}|b_{n-1}\cdots b_0\rangle$ and $i \in \{0,1\}$, measuring qubit b_k outputs "i" with probability

$$P_{k,i} = \sum_{\vec{b}\in\mathbb{Z}_2^n, b_k=i} \left|\alpha_{\vec{b}}\right|^2$$

and the post-measurement state is

$$\frac{1}{\sqrt{P_{k,i}}} \sum_{(b_0,\ldots,b_{k-1},b_{k+1},b_{n-1})\in\mathbb{Z}_2^n} \alpha_{b_{n-1}\cdots b_{k+1}b_{k-1}\cdots b_0}|b_{n-1}\cdots b_{k+1}\, i\, b_{k-1}\cdots b_0\rangle$$

A first application: superdense coding

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  - Principle devised by Bennett and Wiesner in 1970
  - Several experimental validations
- Principle of the protocol: transmit 2 bits of information using a single qubit
  - Alice wishes to send $a_1 a_0 \in \mathbb{Z}_2^2$ to Bob
  - Quantum part: Alice and Bob share a Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B\right)$
  - Alice can apply any transformation to her part of the Bell state before sending it to Bob
  - Bob can then apply any transformation to the Bell state before measuring it

Superdense coding protocol

- On Alice's side
  - She applies X to her part of the Bell state if $a_0 = 1$
  - Then she applies Z if $a_1 = 1$
  - Then she sends her part of the Bell state to Bob

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- On Alice's side
  - She applies X to her part of the Bell state if $a_0 = 1$
  - Then she applies Z if $a_1 = 1$
  - Then she sends her part of the Bell state to Bob
- On Bob's side, upon receiving Alice's qubit
  - He applies $\textsc{cnot}$ to the Bell state
  - Then he applies $H \otimes \mathbf{I}$
  - Then he measures the resulting state

Analysis of Alice's operations

Four cases to consider, with $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $Z \cdot X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

- $a_1 a_0 = 00$: Alice applies nothing to her part of the Bell state before sending it

$$|\psi_{00}\rangle = (\mathbf{I}_A \otimes \mathbf{I}_B) \cdot \left(\frac{1}{\sqrt{2}} \left(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B \right) \right) = \frac{1}{\sqrt{2}} \left(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B \right)$$

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- $a_1 a_0 = 10$: Alice applies Z to her part of the Bell state before sending it

$$|\psi_{10}\rangle = (Z_A \otimes \mathbf{I}_B) \cdot \left(\frac{1}{\sqrt{2}} \left(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B \right) \right) = \frac{1}{\sqrt{2}} \left(|0\rangle_A |0\rangle_B - |1\rangle_A |1\rangle_B \right)$$

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- $a_1 a_0 = 11$: Alice applies $Z \cdot X$ to her part of the Bell state before sending it

$$|\psi_{11}\rangle = ((Z \cdot X)_A \otimes \mathbf{I}_B) \cdot \left(\frac{1}{\sqrt{2}} \left(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B \right) \right) = \frac{1}{\sqrt{2}} \left(|0\rangle_A |1\rangle_B - |1\rangle_A |0\rangle_B \right)$$

Analysis of Bob's operations

Bob receives a state $|\psi_{a_1 a_0}\rangle$ and applies $\textsc{cnot}$ followed by $H \otimes \mathbf{I}$ to it

$$
\begin{aligned}
(H \otimes \mathbf{I}) \cdot \textsc{cnot} \cdot |\psi_{00}\rangle &= (H \otimes \mathbf{I}) \cdot \tfrac{1}{\sqrt{2}} \left(|0\rangle_A |0\rangle_B + |1\rangle_A |0\rangle_B\right) \\
&= (H \otimes \mathbf{I}) \cdot (|+\rangle_A \otimes |0\rangle_B) = |0\rangle_A \otimes |0\rangle_B \\
(H \otimes \mathbf{I}) \cdot \textsc{cnot} \cdot |\psi_{01}\rangle &= (H \otimes \mathbf{I}) \cdot \tfrac{1}{\sqrt{2}} \left(|1\rangle_A |1\rangle_B + |0\rangle_A |1\rangle_B\right) \\
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(H \otimes \mathbf{I}) \cdot \textsc{cnot} \cdot |\psi_{10}\rangle &= (H \otimes \mathbf{I}) \cdot \tfrac{1}{\sqrt{2}} \left(|0\rangle_A |0\rangle_B - |1\rangle_A |0\rangle_B\right) \\
&= (H \otimes \mathbf{I}) \cdot (|-\rangle_A \otimes |0\rangle_B) = |1\rangle_A \otimes |0\rangle_B \\
(H \otimes \mathbf{I}) \cdot \textsc{cnot} \cdot |\psi_{11}\rangle &= (H \otimes \mathbf{I}) \cdot \tfrac{1}{\sqrt{2}} \left(|0\rangle_A |1\rangle_B - |1\rangle_A |1\rangle_B\right) \\
&= (H \otimes \mathbf{I}) \cdot (|-\rangle_A \otimes |1\rangle_B) = |1\rangle_A \otimes |1\rangle_B
\end{aligned}
$$

By measuring the resulting state, he necessarily recovers $a_1 a_0$